Horizon(x) module SMD Argos with RG5804 PA

BGA_RF-RGPA

Table of contents

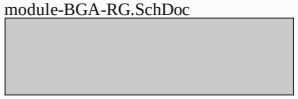
Sheet 1: Project overview (this page)

Sheet 2: module-BGA-RG

Device Sheets:

- Sheet 3: stm32wl55jci7-details
- Sheet 4: STM32RF_power_SMPS
- Sheet 5: CLS_1W_3.3V_Lineup

U_module-BGA-RG
module-BGA-RG.SchDoc



Legend

General comment such as function title, configuration, ...

Text to be added to silkscreen.

Warning text.

Notes to generate the board layout.

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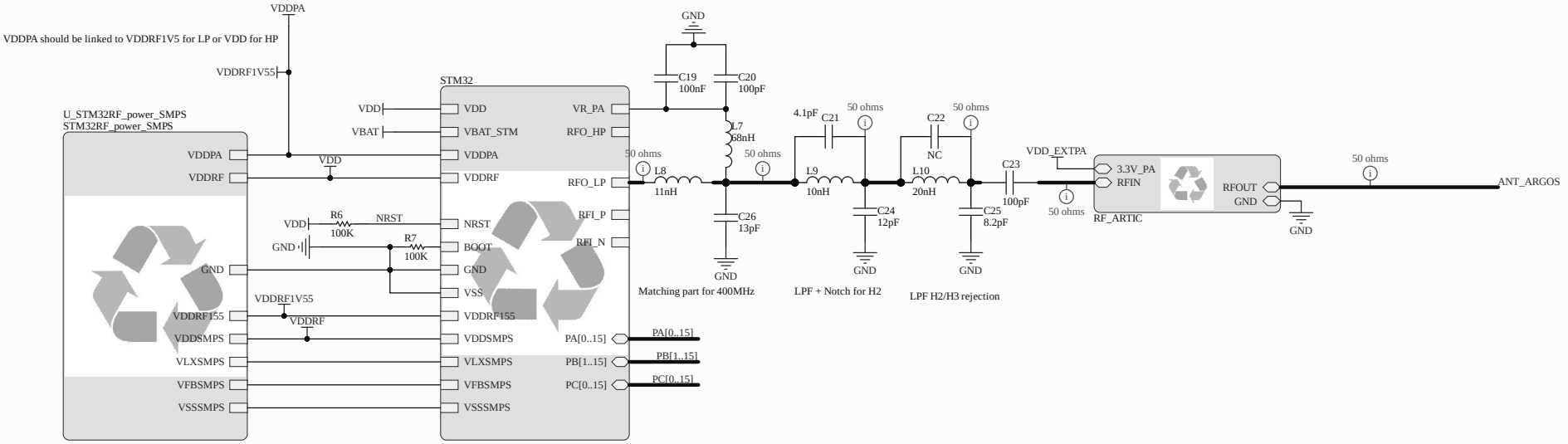
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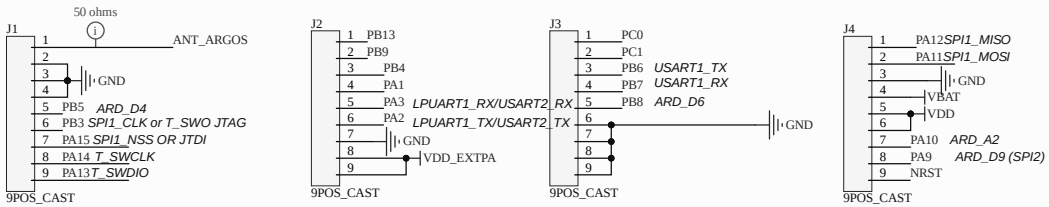
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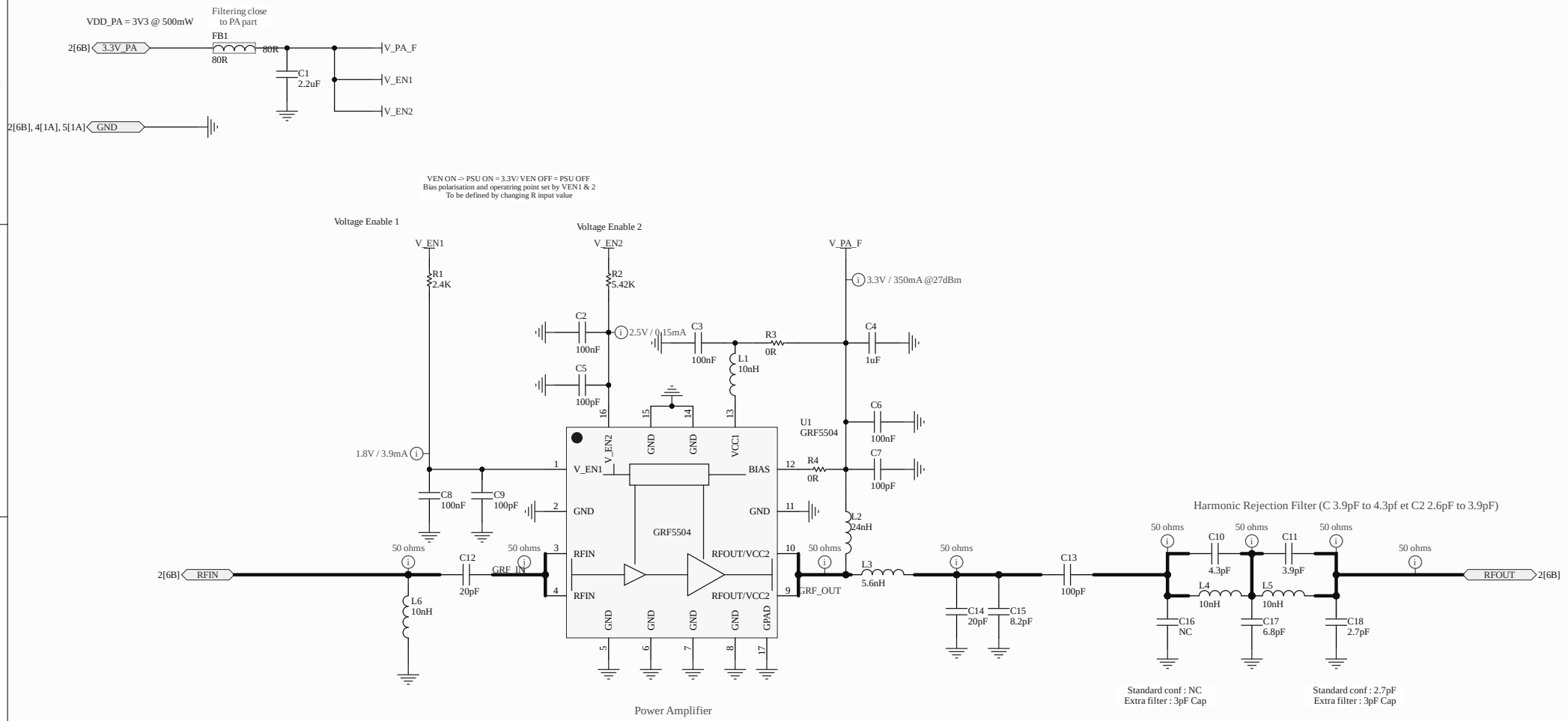
Title: **Schematic sheet title to update**		
Project: **Horizon(x) module SMD Argos with RG5804 PA**		
Variant: [No Variations]		
Revision: -		Reference: BGA_RF-RG
Size: A4	Date:	Sheet: 1 of 5

VDD = 1.8 to 3.6V : External supply for system
VDDSMPS = 1.8V to 3.6V : External supply for SMPS Step down converter
VFBSMPS = 1.45V to 1.62V (1.55v) : External supply main system regulator
VDDA = 0 to 3.6V : 1.71V DAC min, 1.8V buffer, ADC 1.62V VREFBUF = 2.24V
VDDRF = 1.8 to 3.6V :external power supply for radio (VDD)
VDDRF1V = 1.45V to 1.62V for radio
VBAT = 1.55V to 3.6V : RTC Tamp, external clock...
VREF-VREF+: input ref voltage for ADC





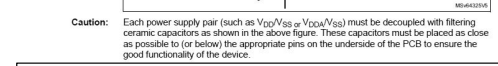
Title		
Size A3	Number	Revision
Date: 8/22/2023		Sheet of
File: D:\Altium\..\module-BGA-RG.SchDoc		Drawn By:



Notes: Murrata
Inductor from LQG ceramic 0402
Inductor from LQW wirewound 0603
Capacitor GRM/GJM 0402

50 ohm Line

Title: ***1W - 3.3V - RF Line-up***			*KINEIS* *11 rue Hermès* *Parc du Canal* *31520 Ramonville St Agne* *France*
Size: A3	Project: BGA_RF-ArticGR.PrjPcb	Revision: V0	
Date: 22/08/2023	Time: 12:40:06	Sheet 3 of 5	
File: CLS_1W_3.3V_Lineup.SchDoc			



Caution: Each power supply pair (such as V_{DD}/V_{SS} or V_{DDA}/V_{SS}) must be decoupled with filtering ceramic capacitors as shown in the above figure. These capacitors must be placed as close as possible to (or below) the appropriate pins on the underside of the PCB to ensure the good functionality of the device.

GND

PA[0..15]

PB[1..15]

PC[0..15]

RF power part output (not managed here)

U2B

VDDSMPS — A2 — VDDSMPS

VLXSMPS — B1 — VLXSMPS

VFBSMPS — B2 — VFBSMPS

VSSSMPS — A1 — VSSSMPS

STM32WL55JCU6

VDD

C35 2.2uF, C36 100nF, C37 100nF, C38 100nF, C39 100nF

VDD Decoupling

close to pins A7, E2, E9, H5

U2D

A7, E2, E9, H5 — VDD

B7, D9, E3, F5 — VSS

VBAT_STM — A8 — VBAT

A5 — VDDA

B5 — VREF+

VBAT decoupling

close to pin A8

C40 1uF, C41 100nF

STM32WL55JCU6

FB2

600R@100MHz

C42 2.2uF, C43 100nF

VDDA decoupling

close to pin A5

U2A

VDDRF155 — F7 — VDDRF1V55

VDDPA — H8 — VDDPA

VDDRF — E8 — VDDRF

VSS — G6, G7, G8, H6 — VSSRF

VR_PA — H9 — VR_PA

RFO_HP — J9 — RFO_HP

RFO_LP — J8 — RFO_LP

RFI_P — J6 — RFI_P

RFI_N — H7 — RFI_N

B1 1000u@100MHz

VDD_TCXO — F6 — PB0-VDD_TCXO

X1

VCC, N.C., OUT, GND

TCXO_OUT — R8 220R

C44 10pF

OSC_IN — G9

OSC_OUT — F8

R9 10K

C45 1uF, C46 100nF

TG2520SMN_32.0000M-MCGNNM0

TCXO 32 MHz 1.8-3.6V 0.5ppm

STM32WL55JCU6

U2E

B1	PA0	D6	PA0	PB1	E7	PB1	ARD_A0
B2	PA1	G3	PA1	PB2	D8	PB2	ARD_A1
LPUART1_TX/USART2_TX	PA2	H2	PA2	PB3	C1	PB3	ARD_D3/T_SWO
LPUART1_RX/USART2_RX	PA3	H1	PA3	PB4	C2	PB4	ARD_A3
ARD_D10	PA4	J1	PA4	PB5	D2	PB5	ARD_D4
ARD_D13	PA5	J2	PA5	PB6	E1	PB6	USART1_TX
ARD_D12	PA6	F4	PA6	PB7	C3	PB7	USART1_RX
ARD_D11	PA7	H3	PA7	PB8	D3	PB8	ARD_D6
MCO	PA8	J3	PA8	PB9	C4	PB9	
ARD_D9	PA9	E5	PA9	PB10	H4	PB10	ARD_D6
ARD_A2	PA10	C8	PA10	PB11	G4	PB11	
ARD_D14	PA11	B9	PA11	PB12	E6	PB12	ARD_D2
ARD_D15	PA12	A9	PA12	PB13	D7	PB13	ARD_A5
T_SWDIO	PA13	B8	PA13	PB14	C6	PB14	ARD_A4
T_SWCLK	PA14	A4	PA14	PB15	B4	PB15	
T_JTDI	PA15	B3	PA15				
	PC0	F2	PC0				
ARD_D7	PC1	F1	PC1				
ARD_D8	PC2	D4	PC2				
	PC3	D5	PC3				
	PC4	F3	PC4				
	PC5	E4	PC5				
B3	PC6	G2	PC6				
	PC13	C7	PC13				

STM32WL55JCU6

U2C

C47 5.1pF, C48 5.1pF

XTAL_IN — B6 — PC14-OSC32_IN

XTAL_OUT — C5 — PC15-OSC32_OUT

PH3-BOOT0 — J5 — BOOT

NRST — F5 — NRST

Y1

NX2012SA-32.768K-STD-MUB-1

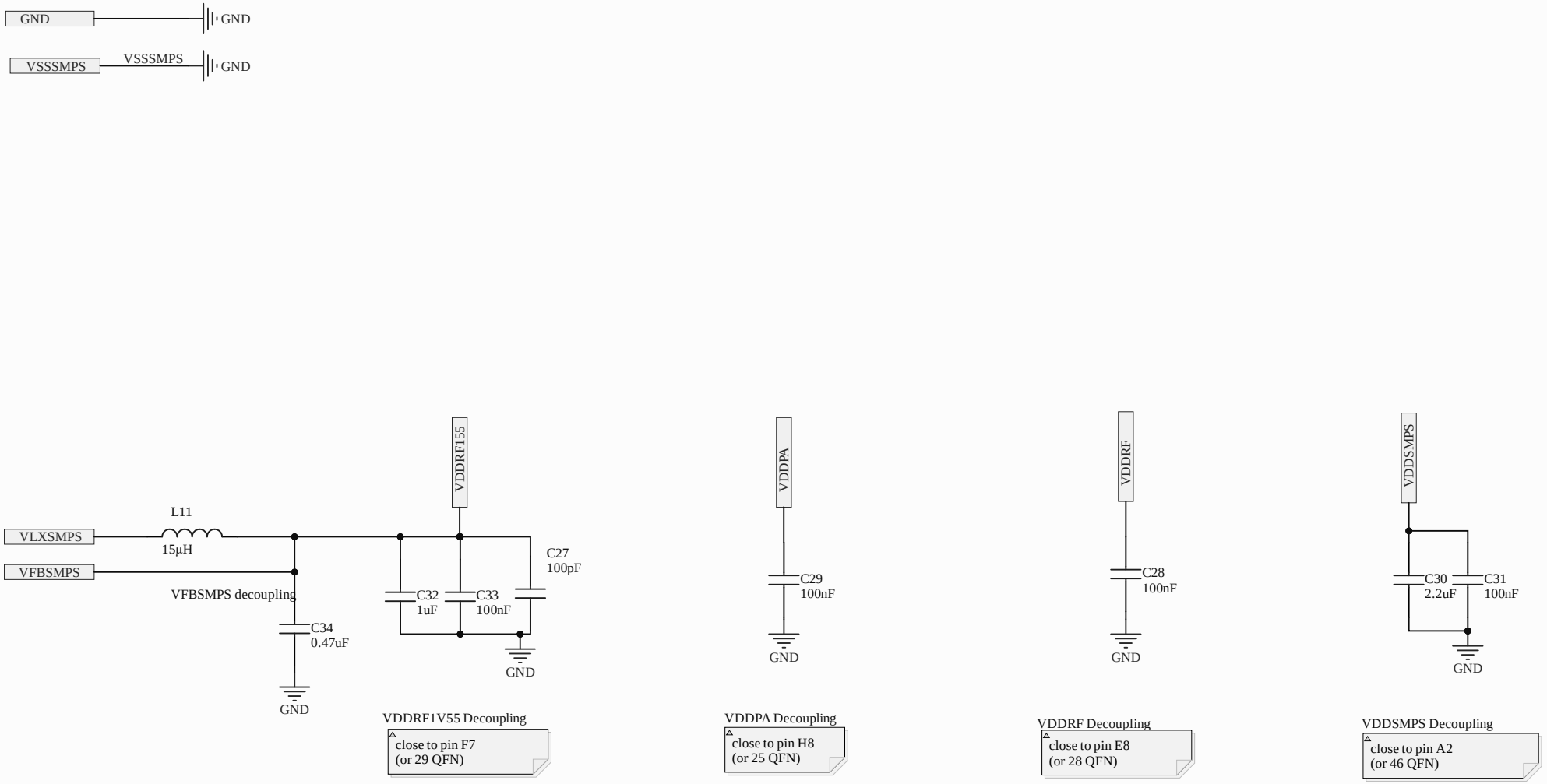
STM32WL55JCU6

C49 100nF

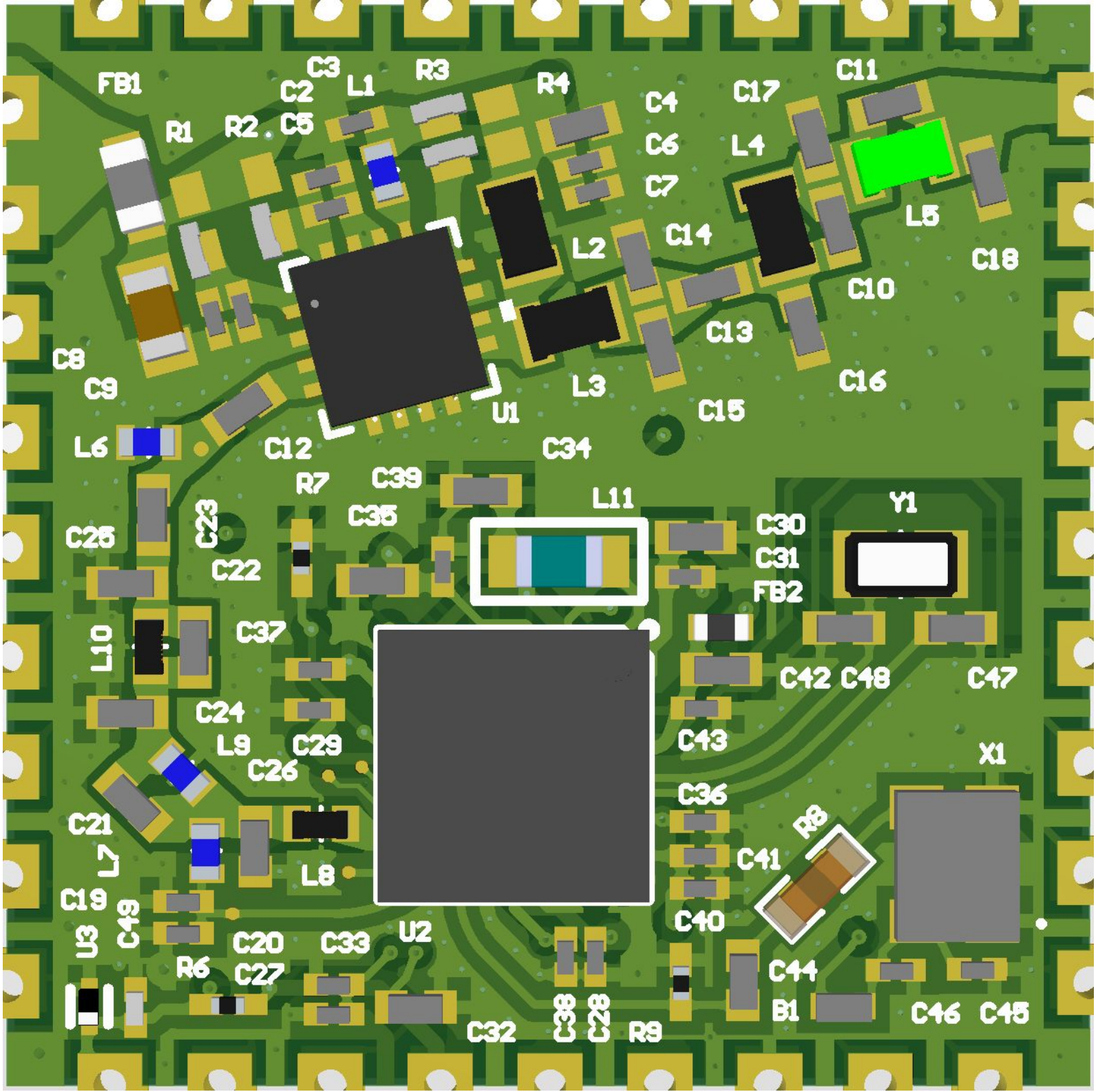
U3 ESDALC6V1-1U2

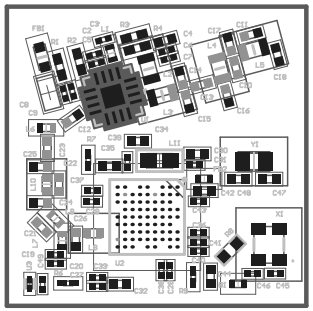
It is highly recommended to apply conformal coatings to the PCB area shown in Figure 13, especially for the LSE quartz, CL1, CL2, and paths to the OSC_IN and OSC_OUT pads as a protection against moisture, dust, humidity, and temperature extremes that may lead to startup problems.

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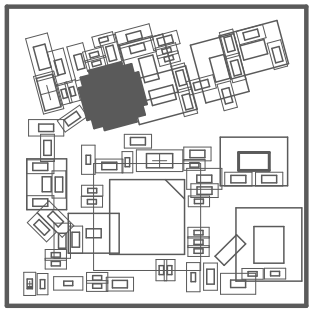


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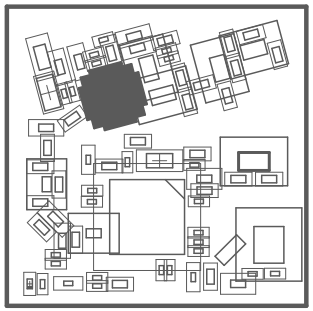

FB1
C3
C2
L1
R3
R4
C4
C17
C11
R1
R2
C5
C6
L4
C7
L5
L2
C14
C18
C10
C8
C13
C9
L3
C16
U1
C15
L6
C12
C34
R7
C39
L11
Y1
C23
C35
C30
C25
C31
C22
FB2
C37
L10
C42
C48
C47
C24
C43
L9
C29
X1
C26
C36
C21
R8
L7
C41
L8
C19
C40
C49
U3
C20
C33
U2
R6
C27
C44
B1
C46
C45
C32
C38
C28
R9



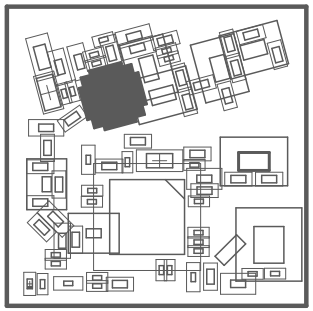
Layer	Name	Material	Thickness	Constant	Board Layer Stack
	Top Overlay				
	Top Solder	Solder Resist	0.020mm	3.5	
1	Top Layer		0.035mm		
	Dielectric 1	1080HR RC68	0.076mm	4.18	
2	Signal Layer 1		0.035mm		
	Dielectric 2	7 x 7628	1.268mm	4.74	
3	Signal Layer 2		0.035mm		
	Dielectric 3	1080HR RC68	0.076mm	4.18	
4	Bottom Layer		0.035mm		
	Bottom Solder	Solder Resist	0.020mm	3.5	
	Bottom Overlay				
Total board thickness:			1.600mm		



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